Learning dynamical systems

Luu Hoang Duc
(MPI MIS & IMH VAST)

Math and Machine Learning series

Contents

Motivation from stochastic calculus

- path-wise RDEs on $(\Omega, \mathcal{F}, \mathbb{P}, (\theta_t)_{t\in R})$: Sell, Oseledets, Millionschikov, Kloeden, Arnold...
$$\dot{y} = f(\theta_t \omega, y). \tag{0.1}$$
- SDEs with Brownian motions: Itô & Stratonovich calculus. Arnold, Kloeden, Caraballo, Schmalfuss, Crauel, Scheutzow...
$$dy_t = f(y_t)dt + g(y_t)dW_t. \tag{0.2}$$
- L.C. Young (1936). An integration of Hölder type, connected with Stieltjes integration. Acta Math.
- K. T. Chen (1954). "Iterated Integrals and Exponential Homomorphisms". Proceedings of the London Mathematical Society.
- H. J. Sussmann (1978). On the gap between deterministic and stochastic ordinary differential equations. Annals of Prob.
- T. Lyons (1998). Differential equations driven by rough signals. Revista Matemtica Iberoamericana.
- SDEs with fractional Brownian motions: rough differential equations. Rough path theory (Lyons 1998), Gubinelli, Friz, Nualart, Hairer...
$$dy_t = f(y_t)dt + g(y_t)dB^H_t, H \in (\frac{1}{4}, 1). \tag{0.3}$$
- Solution Itô-Lyons map: $IL(\omega) = y(T, \omega, y_0)$ of random ODEs and SDEs is measurable w.r.t. ω. But $IL(\omega)$ is continuous w.r.t. ω (Lyons' universal theorem).

Signatures of rough paths

- Let $x \in C^{1-var}(I, \mathbb{R}^d)$. x_t^i is the coordinate i of x_t at time t. Define

$$\mathbf{g}^{k;i_1,\cdots,i_k} := \int_s^t \int_s^{u_k} \ldots \int_s^{u_2} dx_{u_1}^{i_1} \ldots dx_{u_k}^{i_k}.$$

and the set

$$\mathbf{g} = \left(\mathbf{g}^{k;i_1,\cdots,i_k} : 1 \leq k \leq N; i_1, \ldots, i_k \in \{1, \ldots, d\}\right)$$

to be the N-level signature of $x|_{[s,t]}$ and is written as $S_N(x)_{s,t}$. In other words,

$$S_N(x)_{s,t} \equiv \left(1, \int_{s<u<t} dx_u, \ldots, \int_{s<u_1<\ldots<u_k<t} dx_{u_1} \otimes \ldots \otimes dx_{u_k}\right)$$
$$\in \oplus_{k=0}^N \left(\mathbb{R}^d\right)^{\otimes k} =: \mathcal{T}_N(\mathbb{R}^d),$$

- Then $S_N(x)_{s,\cdot}$ satisfies the controlled DE by x as follows

$$\begin{cases} dS_N(x)_{s,t} = S_N(x)_{s,t} \otimes dx_t, \\ S_N(x)_{s,s} = 1. \end{cases}$$

- Rough differential equation

$$dy = V(y)dx \Rightarrow y_{s,t} \approx \tilde{V}(y_s) \otimes S_N(x)_{s,t}$$

- fractional Brownian motion (fBm) with Hurst parameter $H \in (0,1)$: a continuous centered Gaussian process$B^H = (B^H(t), t \in \mathbb{R})$, with covariance function

$$R(s,t) = \frac{1}{2}(|t|^{2H} + |s|^{2H} - |t-s|^{2H}), \qquad s,\ t \in \mathbb{R},$$

When $H = 1/2$, B is the standard Brownian motion.

- B^H is neither Markov nor semi-martingale if $H \neq \frac{1}{2}$!
- Kolmogorov-Centsov theorem: B^H_t is H^- - Hölder continuity a.s. in t.
- For simplicity, think of $dy = g(y)dx$ with scalar path x. First solve the ODE $\dot{u} = g(u)$ to obtain solution $\pi(t, y_0)$ and then assign $y_t = \pi(x_t, y_0)$. Then

$$dy_t = \frac{\partial \pi}{\partial x}(x_t, y_0)dx_t = g(\pi(x_t, y_0))dx_t = g(y_t)dx_t.$$

By Taylor expansion

$$\begin{aligned} y_{a,b} &= \frac{\partial \pi}{\partial x}(x_a, y_0)x_{a,b} + \frac{1}{2}\frac{\partial^2 \pi}{\partial x^2}(x_a, y_0)(x_{a,b})^2 + \ldots \\ &= g(\pi(x_a, y_0))x_{a,b} + \frac{1}{2}\frac{\partial}{\partial x}g(\pi(x_a, y_0))(x_{a,b})^2 + \ldots \\ &= g(y_a)x_{a,b} + \frac{1}{2}Dg(y_a)g(y_a)(x_{a,b})^2 + \ldots \end{aligned}$$

One needs more information from higher order terms if x is of low regularity.

Rough paths

In the simplest form for $\beta \in (\frac{1}{3}, \frac{1}{2})$, a couple $\mathbf{x} = (x, \mathbb{X})$, with $x \in C^{\beta}([a,b], \mathbb{R}^d)$ and $\mathbb{X} \in C_2^{2\beta}([a,b]^2, \mathbb{R}^d \otimes \mathbb{R}^d)$ is called a *rough path* if they satisfy Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \qquad \forall a \leq s \leq u \leq t \leq b. \tag{0.4}$$

We write $\mathbb{X}_{s,t} = (x \otimes x)_{s,t} =: \int_s^t x_{s,u} \otimes dx_u$ and call it Lévy area of x.

- $X_t = \int_0^t a_s dB_s$, we define the stochastic integrals $\int y dX$ as the integral w.r.t. the local martingale X.

$$\mathbb{X}_{s,t} = \int_s^t X_{s,u} dX_u = \frac{1}{2} X_{s,t}^2 - \frac{1}{2}\Big(\langle X \rangle_t - \langle X \rangle_s\Big).$$

- When $X = B^H$ is a fractional Brownian motion, we define the stochastic integral $\int y \delta B^H$ in the sense of Skorohod-Wick-Itô by using the Wick product. Then we can compute explicitly

$$\mathbb{X}_{s,t} \quad := \quad \int_s^t B^H_{s,u} \delta B^H_u = \frac{1}{2}(B^H_{s,t})^2 - \frac{1}{2}\Big(t^{2H} - s^{2H}\Big).$$

Universal approximation theorems

A. Fermanian Computational Statistics and Data Analysis 157 (2021) 107148

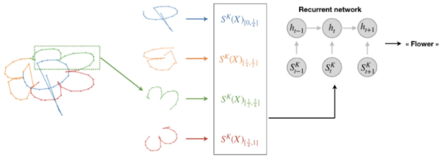


Fig. 3. Signature and recurrent neural network.

Distribution Regression for Sequential Data

Maud Lemercier[1], Cristopher Salvi[2], Theodoros Damoulas[1], Edwin V. Bonilla[3], and Terry Lyons[2]

Theorem 2.1. *Let $\mathcal{X} \subset \mathcal{C}(I, E)$ be a compact set of paths and consider a continuous function $f : \mathcal{X} \to \mathbb{R}$. Then for any $\epsilon > 0$ there exists a truncation level $n \geq 0$ such that for any path $x \in \mathcal{X}$*

$$\left| f(x) - \sum_{k=0}^{n} \sum_{J \in \{1,\ldots,d\}^k} \alpha_J S(x)^J \right| < \epsilon, \qquad (7)$$

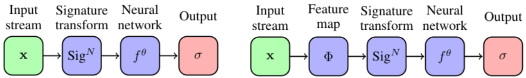
where $\alpha_J \in \mathbb{R}$ are scalar coefficients.

Theorem 3.2. *Let $\mathcal{X} \subset \mathcal{C}(I, E)$ be a compact set of paths and consider a weakly continuous function $F : \mathcal{P}(\mathcal{X}) \to \mathbb{R}$. Then for any $\epsilon > 0$ there exists a truncation level $m \geq 0$ such that for any probability measure $\mu \in \mathcal{P}(\mathcal{X})$*

$$\left| F(\mu) - \sum_{k=0}^{m} \sum_{J \in \{1,\ldots,d\}^k} \alpha_J S(\Phi(\mu))^J \right| < \epsilon, \qquad (14)$$

where $\alpha_J \in \mathbb{R}$ are scalar coefficients.

Signature transform



(a) Neural-signature model.
Trainable parameters: θ.

(b) Neural-signature-augment model.
Trainable parameters: θ.

Figure 1: Two simple architectures with a signature layer.

3.6 Inverting the truncated signature

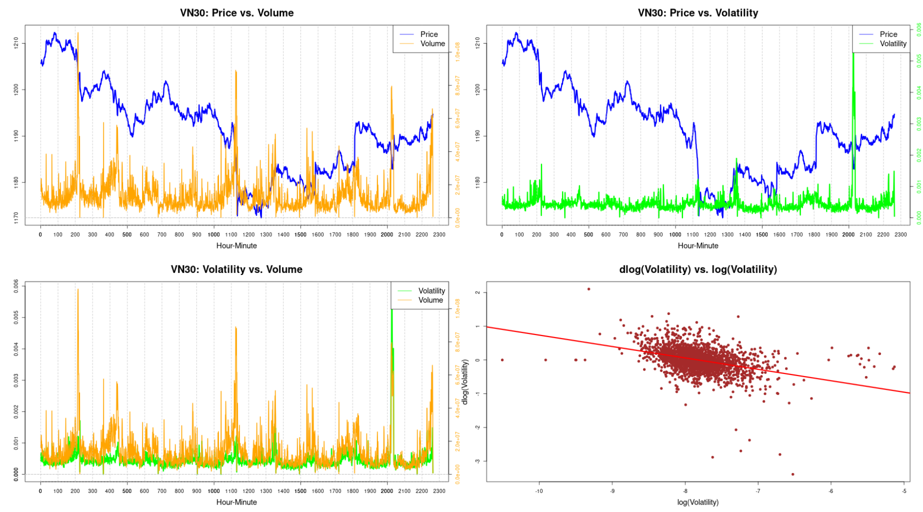
How well does a truncated signature encode the original stream of data? A simple experiment is to attempt to recover the original stream of data given its truncated signature. We remark that finding a mathematical description of this inversion is a challenging task [28, 29, 30].

Fix a stream of data $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{S}(\mathbb{R}^d)$. Assume that the truncated signature $\mathrm{Sig}^N(\mathbf{x})$ and the number of steps $n \in \mathbb{N}$ are known. Now apply gradient descent to minimise

$$L(\mathbf{y}; \mathbf{x}) = \left\|\mathrm{Sig}^N(\mathbf{y}) - \mathrm{Sig}^N(\mathbf{x})\right\|_2^2 \quad \text{for } \mathbf{y} = (y_1, \dots, y_n) \in \mathcal{S}(\mathbb{R}^d).$$

Figure 3 shows four handwritten digits from the PenDigits dataset [31]. The solid blue path is the original path $\mathbf{x}$, whilst the dashed orange path is the reconstructed path $\mathbf{y}$ minimising $L(\mathbf{y}; \mathbf{x})$. Truncated signatures of order $N = 12$ were used for this task. We see that the truncated signatures have managed to encode the input paths $\mathbf{x}$ almost perfectly.

Rough volatility model: VN30 realtime data



Stock price models

1. Samuelson stock model- 1965, Hull-White model - 1987:

$$dS_t = \mu_t S_t dt + \sigma_t S_t dB_t, \tag{1.1}$$

for a stock price S_t, using Itô calculus. σ_t satisfies a stochastic differential equation

$$d \log \sigma_t = k(\theta - \log \sigma_t)dt + \gamma dW_t \tag{1.2}$$

with parameters k, θ, γ and another Brownian motion W_t, with an instantaneous correlation $dB_t dW_t = \rho dt$ with a parameter $\rho \in [0, 1]$ between the two different Brownian motions

2. Heston model - 1993: σ_t^2 follows a Cox-Ingersoll-Ross equation

$$d(\sigma_t^2) = k\Big[\theta - (\sigma_t^2)\Big]dt + \lambda \sigma_t dW_t' \tag{1.3}$$

where $k, \theta, \lambda > 0$ are parameters, and the two Brownian motions again possess an instantaneous correlation $dW_t dW_t' = \rho dt$ with $\rho \in [0, 1]$.

Memory effect

1. Comte-Renault - 1998: ordinary Brownian motion is replaced by fractional Brownian motion in the model (1.2) for the variance σ_t, resulting in

$$d \log \sigma_t = k(\theta - \log \sigma_t)dt + \gamma dB_t^H. \tag{1.4}$$

   In order to obtain a process with long memory, a Hurst exponent $H > \frac{1}{2}$ is needed.

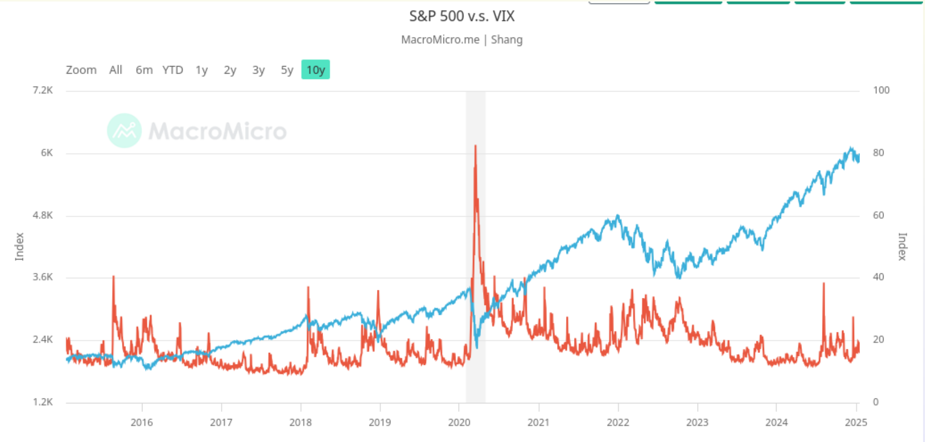
2. Gatheral et al. - (2016,2018): if we assume the model (1.4), the log-volatility behaves essentially as a fractional Brownian motion with Hurst exponent H of order 0.1 at any reasonable time scale. Volatility is rough!

3. Using Hairer's regularity structure theory, Bayer et al. 2019 considers

$$\begin{aligned} \frac{dS_t}{S_t} &= f(Z_t)(\rho dW_t + \sqrt{1-\rho^2} dW'_t) \\ Z_t &= z + \int_0^t K(s,t)v(Z_s)ds + \int_0^t K(s,t)u(Z_s)dW_s, \end{aligned} \tag{1.5}$$

4. Duc-Jost- 2023: consider first equation, using rough path theory (different rough path lift), not Itô calculus. No arbitrage is proved under transaction cost.

Price vs. implied volatility



Long memory and multi-scaling phenomenon

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.5158	0.5103	0.5234	0.5201	0.5278	0.5613	0.5816
Dow Jones	0.5199	0.5177	0.5301	0.5437	0.5534	0.5764	0.5952
Nasdaq100	0.5177	0.5193	0.5245	0.5324	0.5490	0.5694	0.6069

Table: Hurst exponents for different time scales using [R/S] analysis.

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.4898	0.4911	0.4969	0.4838	0.4818	0.5649	0.6006
Dow Jones	0.4937	0.4952	0.5021	0.4927	0.4951	0.5049	0.5386
Nasdaq100	0.4907	0.4964	0.5048	0.5031	0.4755	0.5188	0.5651

Table: Hurst exponents for different time scales using Spectral analysis.

	1M	5M	15M	1H	4H	Daily	Weekly
Sp500	0.5152	0.5202	0.5157	0.5181	0.5149	0.5549	0.5410
Dow Jones	0.5201	0.5208	0.5174	0.5161	0.5081	0.5651	0.5734
Nasdaq100	0.5271	0.5294	0.5224	0.5307	0.5288	0.6112	0.6490

Table: Hurst exponents for different time scales using Higuchi method.

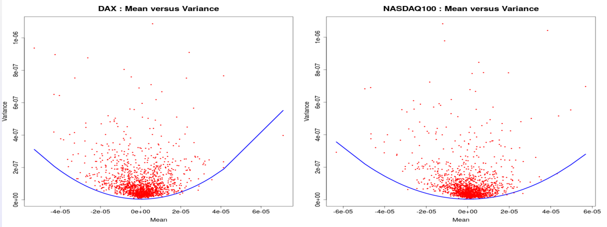
Quadratic relation between variance and expectation of the logarithmic return

A drawback of model (1.1) is the fact that

$$\mathbb{E}Y_{t,t+h} = h\Big(\mu - \frac{\sigma^2}{2}\Big), \quad \text{Var } Y_{t,t+h} = \sigma^2 h,$$

which implies that the variance depends linearly and negatively on the expected return

$$\text{Var } Y_{t,t+h} = 2\mu h - 2\mathbb{E}Y_{t,t+h}. \tag{1.6}$$



Time dependent rough model

- Accordingly, given parameters $\frac{1}{2} > \alpha > \beta > 0$ which also satisfy $2\alpha + \beta > 1$, let us consider the time dependent model
$$dS_t = \mu_t S_t dt + \sigma_t S_t dx_t \tag{1.7}$$
where $\mu : [0, T] \to \mathbb{R}$ is Lebesgue integrable and $\sigma \in C^\beta([0, T], \mathbb{R})$ is controlled by a realization $(\omega, x)^{\mathrm{T}} \in C^\beta([0, T], \mathbb{R}) \oplus C^\alpha([0, T], \mathbb{R})$. Equation (1.7) is understood in the integral form
$$S_t = S_0 + \int_0^t \mu_u S_u du + \int_0^t \sigma_u S_u dx_u. \tag{1.8}$$
- The explicit pathwise solution of (1.7) is given by $S_t = e^{Y_t}$ where
$$Y_b = Y_a + \int_a^b \mu_u du - \frac{1}{2}\int_a^b \sigma_u^2 d[x]_u + \int_a^b \sigma_u dx_u \tag{1.9}$$
- Example: When $X = \hat{B}$ is the G-Brownian motion with respect to the sublinear expectation $\mathbb{E}$, then it follows from Peng that $[\hat{B}] = \langle \hat{B} \rangle$ and one can solve the stochastic differential equation
$$S_t = S_0 + \int_0^t \mu S_u du + \int_0^t \sigma S_u d\hat{B}_u \tag{1.10}$$
explicitly as
$$Y_{s,t} = \mu(t-s) + \sigma \hat{B}_{s,t} - \frac{1}{2}\sigma^2 \Big(\langle \hat{B} \rangle_t - \langle \hat{B} \rangle_s \Big). \tag{1.11}$$

Multi-scaling phenomenon and quadractic relation

$$Y_{t,t+h} = \mu h + \sigma X_{t,t+h} - \frac{\sigma^2}{2}[X]_{t,t+h}, \qquad (1.12)$$

where X and $[X]$ are independent. As a result,

$$EY_{t,t+h} = \mu h - \frac{\sigma^2}{2} E[X]_{t,t+h}, \quad \text{Var } Y_{t,t+h} = \sigma^2 \text{Var } X_{t,t+h} + \frac{\sigma^4}{4} \text{Var } [X]_{t,t+h}. \qquad (1.13)$$

$$\text{Var } X_{t,t+h} = E|X_{t,t+h}|^2 = Ch, \quad \text{Var } [X]_{t,t+h} = C_H h^{2H} \qquad (1.14)$$

It follows from (1.13) that

$$\log\Big(\text{Var } Y_{t,t+h}\Big) = \log\Big(C\sigma^2 h + \frac{C_H}{4}\sigma^4 h^{2H}\Big) \approx \begin{cases} \log(h) + \log(C\sigma^2) & \text{if} \quad h \ll 1 \\ 2H\log(h) + \log(\frac{C_H}{4}\sigma^4) & \text{if} \quad h \gg 1 \end{cases}. \qquad (1.15)$$

$$\text{Var } Y_{t,t+h} = 2\frac{\text{Var } X_{t,t+h}}{E[X,2]_{t,t+h}}\Big(\mu h - EY_{t,t+h}\Big) + \frac{\text{Var } [X,2]_{t,t+h}}{(E[X,2]_{t,t+h})^2}\Big(\mu h - EY_{t,t+h}\Big)^2. \qquad (1.16)$$

In particular, since $\sigma^2 = \frac{2}{E[X,2]_{t,t+h}}\Big(\mu h - EY_{t,t+h}\Big) \geq 0$, it follows that

$$\text{Var } Y_{t,t+h} \geq \frac{\text{Var } [X]_{t,t+h}}{(E[X]_{t,t+h})^2}\Big(\mu h - EY_{t,t+h}\Big)^2$$

$$\Leftrightarrow \quad (EY_{t,t+h}, \text{Var } Y_{t,t+h}) \in \mathcal{P} := \Big\{(x,y) \in \mathbb{R}^2 : y = \frac{\text{Var } [X]_{t,t+h}}{(E[X]_{t,t+h})^2}(\mu h - x)^2\Big\}. \qquad (1.17)$$

Rough volatility models

- Stock model:

$$\begin{aligned} dS_t &= S_t(\mu_t dt + \sigma_t dB_t) \\ \sigma_t &= e^{X_t} \\ dX_t &= \alpha(m - X_t)dt + \nu dB_t^H \end{aligned} \tag{1.18}$$

where B is a standard Brownian motion, B^H is a fractional Brownian motion with Hurst index H such that $cov(B, B^H) = \rho$.

- Data source: realtime data of Vietnam stock index VN30. Order book.
- Discretization scheme

$$\Delta X_t = -\alpha(X_t - m)\Delta t + \nu \Delta W_t^H \tag{1.19}$$

or in the linear form

$$\Delta X_t = c_0 + c_1 X_t + \epsilon_t, \tag{1.20}$$

where

$$c_0 = \alpha m \Delta t, c_1 = -\alpha \Delta t, \epsilon_t = \nu \Delta W_t^H. \tag{1.21}$$

- Hurst index H is estimated using moment method

$$\mathbb{E}[|\Delta W_t^H|^q] \sim (\Delta t)^{qH} \Rightarrow \log \mathbb{E}[|\epsilon_t|^q] = qH \log(\Delta t) \tag{1.22}$$

Learning results: Instability of the model

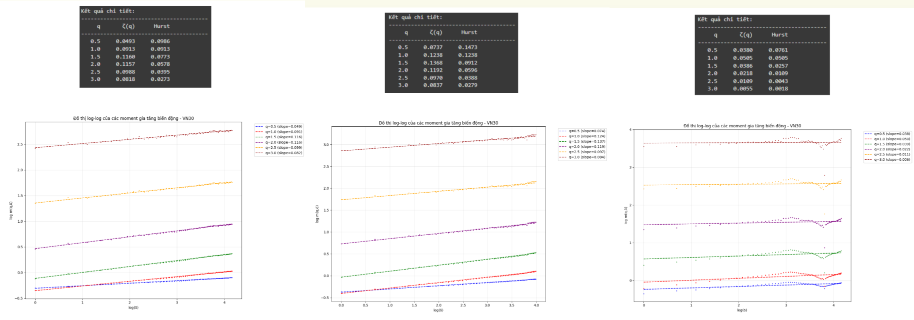
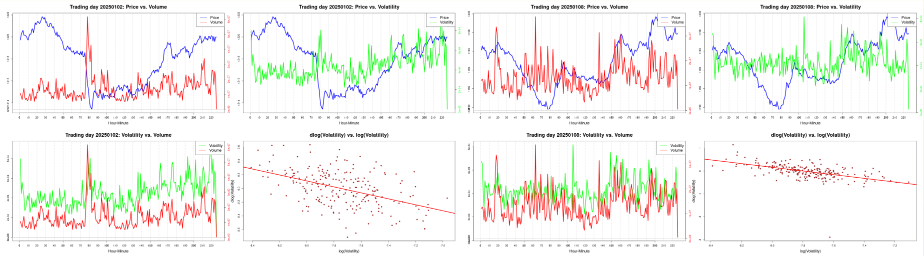
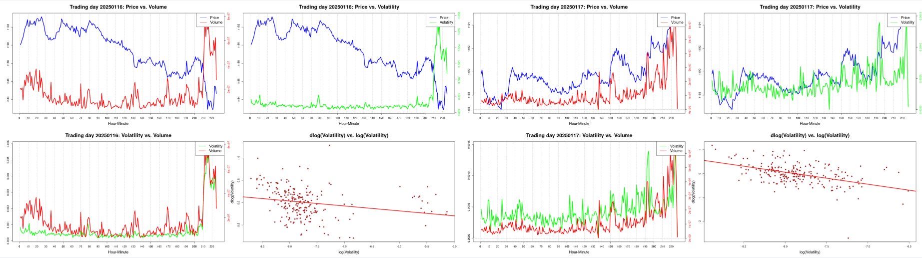


Figure: Different time scale Hurst index

- Instability of the result when time scale is changed.
- When combining data w.r.t. the time scale parameters, it leads to repetition of the data and might create bias.

Learning results: model with trading volume





Price signature method

- The problem is to find $\ell^* \in \mathbb{R}^{d^*}$ such that

$$
\begin{aligned}
\ell^* &\in \arg\min_{\ell} L_{\text{price},\alpha}(\ell), \quad \text{where} \\
L_{\text{price},\alpha}(\ell) &:= \sum_{i=1}^{N} (S_n(\ell) t_i - S_{t_i})^2 + \alpha(\ell) \\
&= \sum_{i=1}^{N} \left(S_0 + \ell_0 \langle e_1, \widehat{\mathbb{X}}_{t_i} \rangle + \sum_{0<|I|\leq n-1} \ell_I \langle \tilde{e}_I, \widehat{\mathbb{X}}_{t_i} \rangle - S_{t_i} \right)^2 + \alpha(\ell),
\end{aligned}
$$

Price signature method

$$\begin{aligned} \mathrm{d}S_t &= \mu(S_t, V_t)\mathrm{d}t + g(S_t, V_t)\mathrm{d}B_t^{\mathbb{P}} \\ \mathrm{d}V_t &= h(S_t, V_t)\mathrm{d}t + \sigma(S_t, V_t)\mathrm{d}W_t^{\mathbb{P}} \end{aligned}$$

with $\mathrm{d}[B^{\mathbb{P}}, W^{\mathbb{P}}]_t = \rho \mathrm{d}t$, where $\rho \in [-1,1]$ and μ, h, g, σ functions from $\mathbb{R}^2$ to $\mathbb{R}$ such that g and σ are strictly positive. Choosing $d = 2$, we aim to estimate the trajectories of a primary process $X_t := (B_t^{\mathbb{Q}}, W_t^{\mathbb{Q}})$ specified by

$$B_t^{\mathbb{Q}} = \int_0^t \frac{\mu(S_s, V_s)}{g(S_s, V_s)}\mathrm{d}s + B_t^{\mathbb{P}}, \qquad W_t^{\mathbb{Q}} = \int_0^t \frac{h(S_s, V_s)}{\sigma(S_s, V_s)}\mathrm{d}s + W_t^{\mathbb{P}}. \tag{4.2}$$

Note that under sufficient integrability conditions on the market prices of risk, $\mathbb{Q}$ is an equivalent local martingale measure, under which $(B^{\mathbb{Q}}, W^{\mathbb{Q}})$ are correlated $\mathbb{Q}$-Brownian motions and under which the dynamics of (S, V) are of the following form:

$$\begin{aligned} \mathrm{d}S_t &= g(S_t, V_t)\mathrm{d}B_t^{\mathbb{Q}}, \\ \mathrm{d}V_t &= \sigma(S_t, V_t)\mathrm{d}W_t^{\mathbb{Q}}. \end{aligned}$$

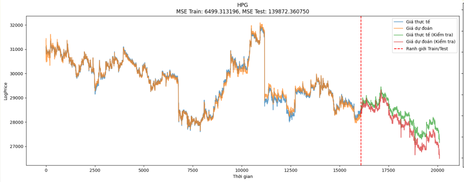
Let us remark that here it is actually easier to work with the market correlated Brownian motion instead of independent ones, as one can avoid explicitly estimating the correlation

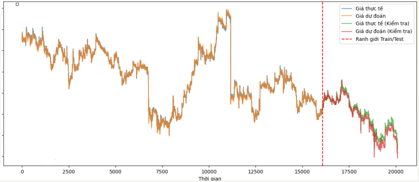
spot covariance estimators in general semimartingale contexts), to get an estimator of $\Sigma_{11} := g(S_t, V_t)^2$ and $\Sigma_{22} := \sigma(S_t, V_t)^2$, which in turn allows computing

$$\mathrm{d}B_t^{\mathbb{Q}} = \frac{\mathrm{d}S_t}{\sqrt{\Sigma_{11,t}}}, \qquad \mathrm{d}W_t^{\mathbb{Q}} = \frac{\mathrm{d}V_t}{\sqrt{\Sigma_{22,t}}}. \tag{4.3}$$

Observe that assuming g and σ to be strictly positive we get that $\sqrt{\Sigma_{11,t}} = g(S_t, V_t)$ and $\sqrt{\Sigma_{22,t}} = h(S_t, V_t)$. This is important to guarantee that $\mathrm{d}S_t = g(S_t, V_t)\mathrm{d}B_t^{\mathbb{Q}}$ (as requested in Corollary 3.12) and $\mathrm{d}V_t = h(S_t, V_t)\mathrm{d}W_t^{\mathbb{Q}}$.

Price signatures method





- Rough path provides more infos but also more constrains that are hard to match.

$$y_{t_0} \in \mathbb{R}^d, \quad y_{t_{k+1}} = y_{t_k} + f(y_{t_k})(t_{k+1} - t_k) + F_{t_k, t_{k+1}}(y, \mathbf{x}), \tag{1.23}$$

$$F_{s,t}(y, \mathbf{x}) = \begin{cases} g(y_s)x_{s,t}, & \mathbf{x} = x \\ g(y_s)x_{s,t} + Dg(y_s)g(y_s)\mathbb{X}_{s,t}, & \mathbf{x} = (x, \mathbb{X}) \end{cases} \tag{1.24}$$

- Discrete systems need not be a discretization scheme of the old form of continuous limiting equations. Probably a new form!

$$\begin{aligned} \varphi^\Delta(0,\omega)y_0 &= y_0, \\ \varphi^\Delta(k\Delta,\omega)y_0 &:= H^\Delta(\theta_{(k-1)\Delta}\omega) \circ \ldots \circ H^\Delta(\omega)y_0, \quad \forall k \geq 1. \end{aligned} \tag{1.25}$$

Suggestion: nonlinear model using RNN

- Need to replace the volatility model by a nonlinear model

$$dX_t = f(X_t)dt + \nu dB_t^H \tag{1.26}$$

  or in the discretization scheme

$$\Delta X_t = f(X_t)\Delta t + \nu \Delta W_t^H \tag{1.27}$$

- f is learnt using RNN. The residual error ϵ_t is used to learn H.
- Or the model for σ is wrong. σ might also depend on another factor like the trading volume.
- Duc-Jost (SIAM Math. Finance 2023): for $B \perp B^H$

$$\begin{aligned} dS_t &= S_t(\mu_t dt + \sigma_t dB_t) \\ d\sigma_t &= \sigma_t(\rho dB_t + \frac{\sqrt{1-\rho^2}dB_t^H}{1+e^{\sqrt{1-\rho^2}B_t^H}}). \end{aligned} \tag{1.28}$$

  Solving using different rough path lifts (W, B) where $dW_t = \alpha(1 - W_t)dB_t^H$ or $W_t = h(B^H) = \frac{e^{\sqrt{1-\rho^2}B_t^H}}{1+e^{\sqrt{1-\rho^2}B_t^H}}$. Then $\sigma_t = W_t e^{\rho B_t}$.

Random dynamical systems

A random dynamical system $\varphi : \mathbb{R} \times \Omega \times \mathbb{R}^2 \to \mathbb{R}^2$, $(t, \omega, y_0) \mapsto \varphi(t, \omega) y_0$ is a measurable mapping w.r.t. (t, ω, y_0) which satisfies the cocycle property

$$\varphi(t + s, \omega) y_0 = \varphi(t, \theta_s \omega) \circ \varphi(s, \omega) y_0, \quad \forall t, s \in \mathbb{R}, \omega \in \Omega, y_0 \in \mathbb{R}^2. \tag{2.1}$$

Generation of an RDS from rough differential equations requires probabilistic setting. Denote by $T_1^2(\mathbb{R}^m) = 1 \oplus \mathbb{R}^m \oplus (\mathbb{R}^m \otimes \mathbb{R}^m)$ the set with the group product

$$(1, g^1, g^2) \bullet (1, h^1, h^2) = (1, g^1 + h^1, g^1 \otimes h^1 + g^2 + h^2),$$

for all $\mathbf{g} = (1, g^1, g^2), \mathbf{h} = (1, h^1, h^2) \in T_1^2(\mathbb{R}^m)$. Then it can be shown that $(T_1^2(\mathbb{R}^m), \bullet)$ is a topological group.

Given $\alpha \in (\frac{1}{3}, \nu)$, assign $\Omega := \mathcal{C}_0^{0,\alpha}(\mathbb{R}, T_1^2(\mathbb{R}^m))$ and equip it with the Borel σ-algebra $\mathcal{F}$. Denote by θ the *Wiener-type shift*

$$(\theta_t \omega)_\cdot = \omega_t^{-1} \bullet \omega_{t+\cdot}, \forall t \in \mathbb{R}, \omega \in \mathcal{C}_0^{0,\alpha}(\mathbb{R}, T_1^2(\mathbb{R}^m)), \tag{2.2}$$

and define the so-called *diagonal process* $\mathbf{X} : \mathbb{R} \times \Omega \to T_1^2(\mathbb{R}^m), \mathbf{X}_t(\omega) = \omega_t$ for all $t \in \mathbb{R}, \omega \in \Omega$. $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space equipped with the continuous (thus measurable) *metric dynamical system* $\theta : \mathbb{R} \times \Omega \to \Omega$. In particular, the Wiener shift (2.2) implies that

$$\|\mathbf{x}(\theta_h \omega)\|_{p-\mathrm{var},[s,t]} = \|\mathbf{x}(\omega)\|_{p-\mathrm{var},[s+h,t+h]}, \quad N_{[s,t]}(\mathbf{x}(\theta_h \omega)) = N_{[s+h,t+h]}(\mathbf{x}(\omega)). \tag{2.3}$$

Numerics for RDE

Discrete time set $\Pi = \{t_k\}_{k\in\mathcal{N}}$ defined by

$$y_{t_0} \in \mathbb{R}^d, \quad y_{t_{k+1}} = y_{t_k} + f(y_{t_k})(t_{k+1} - t_k) + F_{t_k,t_{k+1}}(y,\mathbf{x}), \tag{2.4}$$

$$F_{s,t}(y,\mathbf{x}) = \begin{cases} g(y_s)x_{s,t}, & \mathbf{x} = x \\ g(y_s)x_{s,t} + Dg(y_s)g(y_s)\mathbb{X}_{s,t}, & \mathbf{x} = (x,\mathbb{X}) \end{cases} \tag{2.5}$$

($\mathbf{H_f}$) $f \in C(\mathbb{R}^d,\mathbb{R}^d)$ of linear growth, i.e.

$$\exists C_f > 0 : \quad \|f(y)\| \leq C_f\|y\| + \|f(0)\|, \quad \forall y \in \mathbb{R}^d; \tag{2.6}$$

($\mathbf{H}_g$) either $g \in C^2_b(\mathbb{R}^d,\mathbb{R}^{d\times m})$ such that

$$\|g\|_\infty, C_g := \max\left\{\|Dg\|_\infty, \|D^2g\|_\infty\right\} < \infty, \tag{2.7}$$

or $g(y) = Cy + g(0)$, $C \in (\mathbb{R}^d,\mathbb{R}^{d\times m})$ s.t. $\|C\| \leq C_g$. For YDE, $g \in C^1$ with bounded derivative.

Euler scheme

Theorem (LHD, Kloeden -2023)

Consider the RDE

$$d\hat{y}_t = f(\hat{y}_t)dt + g(\hat{y}_t)dX_t, \hat{y}_a \in \mathbb{R}^d, t \in [a,b] \tag{2.8}$$

where X is a stochastic process with stationary increment which almost all realizations x are of local Hölder continuous with exponent in $(1/3, 1/2]$ such that x is truly rough.
If in additional to ($\mathbf{H_f}$) , ($\mathbf{H_g}$), f and D^2g are local Lipchitz, then there exists a generic random variable $\hat{\xi}(\hat{y}_a, \mathbf{x}, [a,b])$ such that the following estimate holds

$$\begin{aligned}
& \sup_{0\leq k\leq m} \|\hat{y}(t_k, y_a, x) - y_{t_k}\| & (2.9)\\
\leq\ & \hat{\xi}(\hat{y}_a, \mathbf{x}, [a,b])(|\Pi[a,b]| + \|\!|\mathbf{x}|\!\|_{p,[a,b],|\Pi[a,b]|})^{3-p}(b-a+\|\!|\mathbf{x}|\!\|^p_{p,[a,b]}).
\end{aligned}$$

In particular,

$$\lim_{|\Pi[a,b]|\to 0} \sup_{0\leq k\leq m} \|\hat{y}(t_k, \hat{y}_a, x) - y_{t_k}\| = 0. \tag{2.10}$$

Discrete RDS

- Continuous time: Bailleul-Riedel-Scheutzow (2017) proves that RDE $dy_t = f(y_t)dt + g(y_t)dx_t$ generates a continuous RDS $\varphi(t,\omega,y_0)$ on $\mathbb{R}^d$ which is the solution of the RDE w.r.t. the driving path $x(\omega)$ and at initial point y_0.
- Discrete time: Define the discrete mapping $H^\Delta(\mathbf{x})y = y + f(y)\Delta + F_{0,\Delta}(y,\mathbf{x})$. In LHD-Kloeden (2023), we prove that the discrete Euler scheme (2.4) generates a discrete random dynamical system $\varphi^\Delta : \Pi^\Delta \times \Omega \times \mathbb{R}^d \to \mathbb{R}^d$ over $(\Omega,\mathcal{F},\mathbb{P},\theta)$ such that

$$\begin{aligned}\varphi^\Delta(0,\omega)y_0 &= y_0,\\ \varphi^\Delta(k\Delta,\omega)y_0 &:= H^\Delta(\theta_{(k-1)\Delta}\omega)\circ\ldots\circ H^\Delta(\omega)y_0, \quad \forall k\geq 1.\end{aligned} \tag{2.11}$$

- Throughout this paper, we assume that θ is an ergodic dynamical system. In LHD-PT.Hong-ND.Cong (2024), we prove the ergodicity for fBm B^H.

Random pullback attractors

Next, recall that a set $\hat{M} := \{M(\omega)\}_{\omega\in\Omega}$ is called a *random set*, if $\omega \mapsto d(y|M(\omega)) := \inf\{d(y,z)|z \in M(\omega)\}$ is $\mathcal{F}$-measurable for each $y \in \mathbb{R}^d$. A *universe* $\mathcal{D}$ is a family of random sets which is closed w.r.t. inclusions (i.e. if $\hat{D}_1 \in \mathcal{D}$ and $\hat{D}_2 \subset \hat{D}_1$ then $\hat{D}_2 \in \mathcal{D}$).
In our situation, we define the universe $\mathcal{D}$ to be a family of *tempered* random sets $D(\omega)$, that is, $D(\omega)$ is contained in a ball $B(0,\rho(\omega))$ a.s., where the radius $\rho(\omega) > 0$ is a tempered random variable (i.e. $\lim\limits_{t\to\pm\infty} \frac{1}{t}\log^+ \rho(\theta_t\omega) = 0$ a.s.
A random set $\mathcal{A}$ is said to be *invariant* if $\varphi(t,\omega)\mathcal{A}(\omega) = \mathcal{A}(\theta_t\omega)$ for all $t \in \mathbb{R},\ \omega \in \Omega$. An invariant random compact set $\mathcal{A} \in \mathcal{D}$ is called a *pullback attractor* in $\mathcal{D}$, if

$$\lim_{t\to\infty} d_H(\varphi(t,\theta_{-t}\omega)\hat{D}(\theta_{-t}\omega)|\mathcal{A}(\omega)) = 0, \tag{2.12}$$

where $d_H(\cdot|\cdot)$ is the Hausdorff semi-distance, i.e. $d_H(D|A) := \sup_{d\in D}\inf_{a\in A}\|d-a\|$.
A random set $\mathcal{B}^* \in \mathcal{D}$ is called *pullback absorbing* in the universe $\mathcal{D}$ if $\mathcal{B}^*$ absorbs all closed random sets in $\mathcal{D}$, i.e. for any closed random set $\hat{D} \in \mathcal{D}$, there exists a time $t_0 = t_0(\omega,\hat{D})$ such that

$$\varphi(t,\theta_{-t}\omega)\hat{D}(\theta_{-t}\omega) \subseteq \mathcal{B}^*(\omega), \text{ for all } t \geq t_0. \tag{2.13}$$

Then given the universe $\mathcal{D}$ and a compact pullback absorbing set $\mathcal{B}^* \in \mathcal{D}$, there exists a unique pullback attractor $\mathcal{A}(\omega)$ in $\mathcal{D}$, given by

$$\mathcal{A}(\omega) = \bigcap_{t\geq 0} \overline{\bigcup_{s\geq t} \varphi(s,\theta_{-s}\omega)\mathcal{B}^*(\theta_{-s}\omega)}. \tag{2.14}$$

Existence of random attractors

Lemma

Assign

$$\bar{R}_\lambda(\omega) := \sum_{k=0}^{\infty} e^{-k\delta} H(\|\mathbf{x}(\theta_{-k}\omega)\|_{p-\mathrm{var},[-1,0]}) = \sum_{k=0}^{\infty} e^{-k\delta} H(\|\mathbf{x}(\omega)\|_{p-\mathrm{var},[-1-k,-k]}). \qquad (2.15)$$

Then $\bar{R}_\lambda(\cdot)$ *are finite a.s. such that* $\bar{R}_\lambda \in^1$.

Theorem

There exists a pullback attractor $\mathcal{A}$ *for the generated random dynamical system* φ *of the stochastic system such that* $|\mathcal{A}(\cdot)| \in^1$. *Moreover,*

$$\mathcal{A}(\omega) = \bigcap_{n=1}^{\infty} \varphi(n,\theta_{-n}\omega)\bar{\mathcal{B}}^\varepsilon(\theta_{-n}\omega) = \lim_{n\to\infty} \varphi(n,\theta_{-n}\omega)\bar{\mathcal{B}}^\varepsilon(\theta_{-n}\omega), \quad \forall \omega \in \Omega. \qquad (2.16)$$

where $\bar{\mathcal{B}}^\varepsilon$ *is given by*

$$\bar{\mathcal{B}}^\varepsilon(\omega) := V^{-1}(K_\lambda + \bar{R}_\lambda(\omega) + \varepsilon), \quad \forall \varepsilon \geq 0. \qquad (2.17)$$

Existence of continuous attractors

Now we assume further that f is dissipative: there exist $c, d > 0$ so that for all $y \in \mathbb{R}^d$,

$$\langle y, f(y) \rangle \leq c - d\|y\|^2. \tag{2.18}$$

In addition, assume the p-variation norm of $\mathbf{x}$ is of finite moment for all order $k \in \mathcal{N}$, i.e.

$$\mathbb{E}\, \|\mathbf{x}(\cdot)\|^k_{p,\Pi[a,b]} < \infty, \quad \forall k \in \mathcal{N}, \quad \forall 0 < a < b, \ a, b \in \Pi. \tag{2.19}$$

Theorem (LHD-2022)

Assume $(\boldsymbol{H}'_f), (\boldsymbol{H}^b_g), (\boldsymbol{H}_X)$, then there exists a random pullback attractor $\mathcal{A}$ such that $|\mathcal{A}(\cdot)| \in^{\rho}$ for any $\rho \geq 1$. If in addition $(\boldsymbol{H}_{\mathcal{A}})$ and f is global Lipschitz continuous then the random attractor is upper semi-continuous with respect to the noise intensity in the sense that $\mathcal{A} \to \mathcal{A}$ a.s. (w.r.t. the Hausdorff semi-distance) as $C_g \to 0$, both in the almost sure and in $\mathcal{L}^{\rho}$ senses. Moreover, if f is strictly dissipative then $\mathcal{A}$ is a singleton provided that C_g is sufficiently small.

Semi-continuity of attractors

Theorem (Cong-LHD-Hong-2022)

Assume ($\mathbf{H_f}$), ($\mathbf{H_g}$) *and the dissipativity condition* (2.37) *and condition* (2.19). *Consider system* (2.4) *with the regular time set* Π^Δ, *where* $\Delta \in (0,1)$ *satisfied the inequality*

$$0 < \Delta < 1 \wedge \frac{d}{2C_f^2} \wedge \frac{1}{2d}. \tag{2.20}$$

Then the generated discrete random dynamical system φ^Δ *of* (2.4) *admits a random pullback attractor* $^\Delta$.

Theorem

Under assumptions ($\mathbf{H_f}$), ($\mathbf{H_g}$) *and for all* Δ *small enough, the attractor* $\mathcal{A}^\Delta$ *and the absorbing set* $\mathcal{B}^\Delta$ *of* (2.4) *is uniformly bounded in* Δ *a.s.*
As a result, for L_g *small enough, the numeric attractor* $\mathcal{A}^\Delta$ *converges to the attractor* $\mathcal{A}$ *in the Hausdorf semi-distance, i.e.* $d_H(\mathcal{A}^\Delta|\mathcal{A}) \to 0$ *as* $\Delta \to 0$, *a.s.*

RDEs and Doss-Sussmann technique

Solve the controlled differential equation

$$dy_t = f(y_t)dt + g(y_t)dx_t, \tag{2.21}$$

we can also interpret equation (2.21) in the integral form

$$y_t = y_0 + \int_0^t f(y_s)ds + \int_0^t g(y_s)dx_s, \quad \forall t \geq 0, \tag{2.22}$$

where the second integral is a rough integral for the rough path y *controlled* by x in the sense of Gubinelli. The techniques to be used are the Doss-Sussmann technique, i.e. solution $\phi_\cdot(\mathbf{x}, \phi_a)$ of the *pure* rough differential equation

$$d\phi_u = g(\phi_u)dx_u, \quad u \in [a,b], \phi_a \in \mathbb{R}^d \tag{2.23}$$

is C^1 w.r.t. ϕ_a, and $\frac{\partial \phi}{\partial \phi_a}(\cdot, \mathbf{x}, \phi_a)$ is the solution of the linearized system

$$d\xi_u = Dg(\phi_u(\mathbf{x}, \phi_s))\xi_u dx_u, \quad u \in [a,b], \xi_a = Id, \tag{2.24}$$

where $Id \in \mathbb{R}^{d\times d}$ denotes the identity matrix. The Doss-Sussmann transformation $y_t = \phi_t(\mathbf{x}, z_t)$ map solution y_t of (2.21) on a certain interval $[0,\tau]$ to solution z_t of the associate ordinary differential equation

$$\dot{z}_t = \Big[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t)\Big]^{-1} f(\phi_t(\mathbf{x}, z_t)), \quad t \in [0,\tau],\ z_0 = y_0. \tag{2.25}$$

Doss-Sussmann technique

$$\begin{aligned}\left\|\!\left|\phi\right|\!\right\|_{p-\mathrm{var},[a,b]} &\leq 8C_pC_g \left\|\!\left|\mathbf{x}\right|\!\right\|_{p-\mathrm{var},[a,b]};\\ \left\|\frac{\partial \phi}{\partial \phi_a}(t,\mathbf{x},\phi_a) - Id\right\|, \left\|\Big[\frac{\partial \phi}{\partial \phi_a}(t,\mathbf{x},\phi_a)\Big]^{-1} - Id\right\| &\leq 16 C_pC_g \left\|\!\left|\mathbf{x}\right|\!\right\|_{p-\mathrm{var},[a,b]}.\end{aligned} \tag{2.26}$$

Introduce the notations

$$\eta_t := y_t - z_t, \quad \text{and} \quad \psi_t := \Big[\frac{\partial \phi}{\partial z}(t,\mathbf{x},z_t)\Big]^{-1} - Id, \quad \forall t \in [0,\tau],$$

where $\tau > 0$ is chosen such that $16C_pC_g \left\|\!\left|\mathbf{x}\right|\!\right\|_{p-\mathrm{var},[0,\tau]} \leq \lambda$ for some $\lambda \in (0,1)$ small enough. With such τ, it then follows from (2.26) that

$$\|\eta_t\| = \|\phi_t(\mathbf{x},z_t) - z_t\| \leq \frac{\lambda}{2} \quad \text{and} \quad \|\psi_t\| \leq \lambda, \quad \forall t \in [0,\tau]. \tag{2.27}$$

To estimate $\|z_t\|$, we rewrite (2.25) as

$$\dot{z}_t = (Id + \psi_t) f(z_t + \eta_t). \tag{2.28}$$

Lyapunov functions

($\mathbf{H}_V$) There exists for a Lyapunov-type function V for the drift f which satisfies

- there exists positive definite functions $\alpha, \beta > 0$ such that

$$\alpha(\|z\|) \leq V(z) \leq \beta(\|z\|), \quad \forall z \in \mathbb{R}^d; \tag{2.29}$$

- there exists a constant $L_V > 0$ such that

$$\|\nabla V(z)\| \leq L_V, \quad \forall z \in \mathbb{R}^d; \tag{2.30}$$

- there exists a sufficiently small constant $\lambda \in (0,1)$ and constants $C_\lambda, \delta > 0$ such that

$$\sup_{\|\psi\|,\|\eta\| \leq \lambda} \langle \nabla V(z), (I+\psi) f(z+\eta) \rangle \leq C_\lambda - \delta V(z), \quad \forall z \in \mathbb{R}^d. \tag{2.31}$$

Condition (2.31) is stronger than the classical condition on the gradient ∇V along the vector field f: there exist constants $d_1, d_2 > 0$ such that

$$\langle \nabla V(z), f(z) \rangle \leq d_1 - d_2 V(z), \quad \forall z \in \mathbb{R}^d. \tag{2.32}$$

(2.31) is designed in such a way that it does not depend on g but only on the control parameter λ that relates to the intensity C_g.

Stability with Lyapunov functions

Theorem

Under the assumptions $(\boldsymbol{H}_V)$, $(\boldsymbol{H}_g^b)$ *and for any realization* x *satisfying* $(\boldsymbol{H}_X)$, *there exists a solution of* (2.21) *on any interval* $[0,T]$. *Moreover, there exist a constant* $\delta > 0$ *such that any sufficiently small* $\lambda \in (0,1)$ *is associated with a constant* $K_\lambda > 0$ *of which the following estimate holds*

$$V(y_t) - K_\lambda \le e^{-\delta t}\Big[V(y_0) - K_\lambda\Big] + L_V \lambda \int_0^t e^{-\delta s} dN^-\Big(\frac{\lambda}{16 C_p C_g}, \mathbf{x}, [t-s,t]\Big), \quad \forall t \in [0,T]. \tag{2.33}$$

Corollary

Under the assumptions of Theorem 7, there exist constants $\delta > 0$ *such that for any* $\lambda \in (0,1)$ *small enough, there exist a constant* $K_\lambda > 0$ *such that the following estimate holds for all*

$$V(y_t) - K_\lambda \le e^{-\delta t}\Big[V(y_0) - K_\lambda\Big] + H(\|\mathbf{x}\|_{p-\mathrm{var},[0,t]}), \quad \forall t \in [0,T] \tag{2.34}$$

where

$$H(\xi) = L_V (16 C_p C_g)^p \lambda^{1-p} \xi^p + 8 L_V C_p C_g \xi. \tag{2.35}$$

Applications

- Globally Lipschitz continuous drifts
$$\|f(z_1) - f(z_2)\| \leq C_f \|z_1 - z_2\|, \quad \forall z_1, z_2 \in \mathbb{R}^d. \tag{2.36}$$
Then condition (2.32) follows condition (2.31).
- Consider f to satisfy the global Lipschitz continuity and the dissipativity condition, i.e. there exists $d_1 \geq 0, d_2 > 0$ such that
$$\langle z, f(z) \rangle \leq d_1 - d_2 \|z\|^2, \quad \forall z \in \mathbb{R}^d. \tag{2.37}$$
Then construct $V(z) := \sqrt{1 + \|z\|^2}$.
- The damped pendulum system in 2 dimension $z = (v, w) \in \mathbb{R}^2$ given by
$$\begin{cases} dv_t &= w_t \\ dw_t &= -\sigma^2 \sin v_t - 2\mu w_t \end{cases} \tag{2.38}$$
Then f is globally Lipschitz continuous. $V(z) := \sqrt{1 + \frac{1}{2} w^2 + \sigma^2 (1 - \cos v)}$.
- Fitzhugh-Nagumo neuro-system
$$\begin{aligned} dv_t &= v_t - \frac{v_t^3}{3} - w_t + I \\ dw_t &= \varepsilon (v_t - \mu w_t + J) \end{aligned} \tag{2.39}$$
Then construct $V(y) := \left(1 + v^4 + \frac{6}{\varepsilon\mu} w^2\right)^{\frac{1}{4}}$.

Stability at equilibrium

For simplicity of presentation, throughout this section we assume that f is globally Lipschitz continuous with $f(0) = 0$, while $g \in C_b^3$ with $g(0) = 0$. Then

$$\|\phi\|_{p-\mathrm{var},[a,b]} \leq 8C_pC_g \|\mathbf{x}\|_{p-\mathrm{var},[a,b]} \min\{\|\phi_a\|, 1\}. \tag{2.40}$$

Given the Doss-Sussmann transformation $y_t = \phi_t(\mathbf{x}, z_t)$, the difference $\eta_t := y_t - z_t$ on each time interval $16C_pC_g \|\mathbf{x}\|_{p-\mathrm{var},[\tau_i,\tau_{i+1}]} \leq \lambda$ then satisfies $\|\eta_t\| \leq \lambda\|z_t\|$ for all $t \in [\tau_i, \tau_{i+1}]$. Moreover η_t is then a Lipschitz continuous function of z_t. This motivates us to define $L(\lambda)$ to be the convex set of Lipschitz continuous function η of z such that $\|\eta(z)\| \leq \lambda\|z\|$.
Therefore, the assumptions for our targeted Lyapunov function V^* are modified with additional conditions that $V^*(0) = 0$ and $C_\lambda = 0$ in (2.31), i.e. there exists $\lambda \in (0, 1)$ sufficiently small and a constant $\mu > 0$

$$\sup_{\|\psi\|\leq\lambda,\eta\in L(\lambda)} \langle \nabla V^*(z), (I + \psi)f(z + \eta(z))\rangle \leq -\mu V^*(z), \quad \forall z \in \mathbb{R}^d. \tag{2.41}$$

Lyapunov neural networks

Lyapunov neural network $V_\vartheta : \mathbb{R}^d \to \mathbb{R}_+$ from an arbitrary neural network $N_\vartheta : \mathbb{R}^d \to \mathbb{R}$ with the set of its κ-trainable parameters $\vartheta \in \mathbb{R}^\kappa$, given by

$$V_\vartheta(z) := |N_\vartheta(z) - N_\vartheta(0)| + \bar{\alpha}\|z\| \tag{2.42}$$

where $\bar{\alpha} > 0$ is a small user-chosen parameters.
Recall that the training process of Lyapunov neural network V_ϑ in (2.42) aims to find a specific network parameter ϑ such that the condition (2.31) holds for V_ϑ at every $z \in \mathbb{D}$ of a compact domain $\mathbb{D}$. This can be achieved by minimize the risk function

$$I(\vartheta) := \frac{1}{|\mathbb{D}|}\int_{\mathbb{D}} \Big[\sup_{\|\psi\|\leq\lambda,\eta\in L(\lambda)} \langle \nabla V_\vartheta(z), (I+\psi)f(z+\eta(z))\rangle + \bar{\mu}V_\theta(z)\Big]_+ dz \tag{2.43}$$

for a user chosen parameter γ. In practice, the integral (2.43) does not have analytic form, thus we approximate $I(\vartheta)$ using the empirical expectation

$$\hat{I}(\vartheta) := \frac{1}{M}\sum_{i=1}^{M} \Big[\sup_{\|\psi\|\leq\lambda,\eta\in L(\lambda)} \langle \nabla V_\vartheta(z_i), (I+\psi)f(z_i+\eta(z_i))\rangle + \bar{\mu}V_\theta(z_i)\Big]_+ \tag{2.44}$$

where $\{z_i : i = 1, \dots, M\}$ are independent and identically distributed (i.i.d) samples from the uniform distribution $U(\mathbb{D})$ on $\mathbb{D}$.

Main Results

Lemma (Universal approximation)

Given $\|h\|_{W^{1,\infty}(\mathbb{D})} < B$ *for some* $B > 0$ *and for all* $\varepsilon \in (0, \frac{1}{2})$*, there exists a feedforward neural network* N_ϑ *with the network parameter* ϑ *and the activation function* $RePU(z) := \max\{0, z\}$ *such that* $\|N_\vartheta - h\|_{W^{1,\infty}(\mathbb{D})} \leq \varepsilon$*, where*

$$\|h\|_{W^{1,\infty}(\mathbb{D})} := \max\{\mathrm{esssup}_{z\in\mathbb{D}}\|h(z)\|, \max_{1\leq i\leq d} \mathrm{esssup}_{z\in\mathbb{D}}\|D_i h(z)\|\}$$

Lemma (Lya Nets are dense)

For any $\delta > 0$*, the set of Lyapunov-Nets of the form* (2.42) *with RePU network* N_ϑ *and bounded weights* $\vartheta \in [-1, 1]^\kappa$ *is dense in the function space* $\mathcal{S} := \{\bar{h}(z) + \alpha\|z\| : \bar{h} \in C^1(\mathbb{D}, \mathbb{R}_+), \bar{h}(0) = 0\}$ *under* $W^{1,\infty}(\mathbb{D}_\delta)$ *where* $\mathbb{D}_\delta = \mathbb{D} \setminus B(0, \delta)$*.*

Lemma

For any $\delta, c \in (0, 1)$*, there exists an integer* $M = M(\delta, c) \in \mathcal{N}$ *such that for some* M *sampling points* $z_i \in \mathbb{D}_\delta$ *where* $i = 1, \ldots, M$ *we obtain* $\mathbb{D}_\delta \subset \bigcup_{i=1}^M B(z_i, c\|z_i\|)$

Main results

Lemma

For any $\varepsilon \in (0, \frac{\mu\alpha}{4C_f+2\mu})$ and $\delta > 0$, there exists $\vartheta^* \in \Theta$ such that V_{ϑ^*} satisfies (2.29) and

$$\sup_{\|\psi\|\leq\lambda,\eta\in L(\lambda)} \langle \nabla V_{\vartheta^*}(z), (I+\psi)f(z+\eta(z))\rangle \leq -a_{\mu,\varepsilon} V_{\vartheta^*}(z), \quad \forall z \in \mathbb{D}_\delta, \tag{2.45}$$

where $a_{\mu,\varepsilon} := \mu - \frac{\varepsilon}{\alpha-\varepsilon}(\mu + 4C_f) > 0$.

For bounded weights Θ and the fixed network V_ϑ, the function $\langle \nabla V_\vartheta(y), (I+\psi)f(y+\eta(y))\rangle$ is also globally Lipschitz continuous w.r.t. y on $\mathbb{D}$, with a global Lipschitz constant $L(\mathbb{D},\Theta)$ that is independent of ϑ.

Theorem

For any $\delta \in (0,1)$ and $a_{\mu,\varepsilon}$ as defined in Lemma 12, choose any $\bar{\gamma} \in (0, a_{\mu,\varepsilon})$ and $0 < c < \frac{\bar{\gamma}(\alpha-\varepsilon)}{L(\mathbb{D},\Theta)} < 1$ (if $\frac{\bar{\gamma}(\alpha-\varepsilon)}{L(\mathbb{D},\Theta)} \geq 1$ then choose any $c \in (0,1)$). Let M and $Z_M := \{z_i : i = 1, \dots M\}$ be given as in Lemma 11 and the empirical risk function $\hat{I}(\vartheta)$ as in (2.44). Then the minimizer of $\hat{I}$ exists and achieves minimal value 0. Moreover, for any minimizer $\hat{\vartheta}$ of $\hat{I}(\cdot)$, the corresponding $V_{\hat{\vartheta}}$ is an δ-accurate Lyapunov function of f on $\mathbb{D}$.

Thank you!